AUtoMLy | Medicaid Policy Copilot

INAI 6986 — Milestone 1: Data and Model Selection

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Decision and Evidence Boundary

This post-feedback update incorporates the October 1 corrected training runs and verified October 8 development review. Sixteen nested-size training runs across four families used the fixed 124 validation prompt IDs and zero protected test/control rows. The proposed Nemotron checkpoint is only a provisional evaluation candidate: a 192-slot advisory probe tied three adapters at 4/24 fully supported answers and was neither human-scored nor a final test. Fresh-source exposure and final-seal gates remain unresolved; no deployment claim is made.

1. Data Characteristics

The task is a source-cited answer to a public Alabama Medicaid policy or provider-billing question, with a source locator or an abstention when support is absent. It is not an automated eligibility, payment, clinical or legal decision. Each candidate retains source URL, document identity, PDF page, snapshot date and SHA-256 file fingerprint. The target is a response grounded in the frozen source, not a class label.

Prepared Evidence	Measured Shape / Consequence
Source snapshot	138 public PDF files / 2,433 pages (September 14); 136 yielded candidate records.
Ordinary examples	1,999 answerable supervised fine-tuning (SFT) records; a separate one-record intentionally unanswerable case was quarantined.
Source mix	750 Administrative Code and 1,249 Provider Manual records. Median answer length 18 whitespace-delimited words; 95th percentile 52.
Historical clean split	1,684 train / 124 validation / 191 historical test records; 113 / 10 / 13 distinct source documents, respectively; zero document overlap.
Rights and privacy	Approved public information only; no patient claims, protected health information or agency-confidential records.

Preparation and Revised Split

The document-grouped split assigns every page of a source PDF together, preventing page-neighbor leakage. One blank-page safety example was removed from ordinary SFT. Earlier September 16–18 Muse experiments used pre-amendment generator, split and training procedures. They are retained in the historical archive for provenance only; none of their run metrics is pooled into the current comparison, tables or plots.

The new experiment nests training subsets of 200, 509, 750 and 1,600 from the same 1,684 clean training rows, covers all 113 training documents even at 200, and holds the same 124 validation rows. Earlier proposals for separate 100/250/200/300-record configuration tests were deferred to protect one final test event. The already exposed historical 191 are retrospective diagnostics, not an untouched final test. The thirty hand-authored control questions remain locked; all their source documents are in the historical test group, none in train or validation. Their question wording and keys are independently authored, but their documents are not independent of the historical test documents.

2. Candidates Considered

The declared family is deep learning: local language-model adaptation. Four pinned model families now have real project-data training at each nested size. Families differ in tokenizer, low-rank adaptation recipe, memory needs, and serving practicality. This is a set of plausible engineering candidates, not a ranking of policy-answer quality. The older Muse A/B render comparison was one recipe plus a sensitivity run, not a second substantive candidate.

Approach	Why Plausible / Constraint	Verified Status
Muse Glimmer 30B; four-bit rank-16 adapter	Local serving baseline and memory-efficient adaptation; format gains alone do not establish grounding.	Four clean N=200/509/750/1,600 runs completed Sep. 29–30; prior-method September 16–18 runs excluded.
Gemma 4 26B-A4B; BF16 rank-4 adapter	Alternative mixture-of-experts architecture; its tokenizer makes token metrics cross-family incomparable.	Four runs complete; N=1,600 best validation checkpoint 400, final step 600.
Nemotron 3.5 Lightning 30B-A3B; BF16 rank-16 adapter	New candidate capacity and architecture; RTX PRO memory and validated full training path are required.	Four corrected-code runs complete Oct. 1; N=1,600 best checkpoint 400, final step 1,200.
Ornith 1.5 35B-A3B; BF16 rank-16 adapter	Separate local architecture and serving trade-offs; no factual claim from token loss.	Four corrected-code runs complete Oct. 1; N=1,600 best checkpoint 400, final step 1,200.

3. Baseline

The practice reference is manual source lookup and answer drafting. Thirty original student-authored code/page lookups were source-validated; paired retrospective session timestamps estimate 42:06 total (about 1:24 each). This is an easy-task proxy, not a stopwatch trial or a representative policy-reasoning baseline. The proposed model-vs-untouched-base comparison is a separate technical control and has not been run on a locked final cohort.

Why the Original Control Cannot Decide the Winner

All 30 items ask an answerable page or code lookup. Ten come from Jul26_34.pdf page 2, six from Jul26_28.pdf page 2 and five from Apr26_37.pdf page 2: 21/30 on three pages. This is an honest lookup diagnostic, not a representative policy-support or abstention benchmark. Nine supplementary pairs were submitted in Batch A (eight adopted local-AI drafts and one edited), not independently authored new controls. The user approved a separate local-AI/source-verified reasoning and unsupported-question benchmark; it has not yet been created. This changes authorship, not the instructor criterion, and no waiver is claimed.

4. Evidence and Evaluation

The intended primary outcome is human-adjudicated factual accuracy on answerable policy questions; source locator, unsupported claims and abstention are safety-critical secondary outcomes. No same-split final comparison or final human-adjudicated model accuracy exists. A post-hoc local advisory v2 development probe scored 24 question-only slots per family and condition (not the fresh-source final benchmark): fully supported counts were Nemotron 4/24 base and 4/24 adapter, Gemma 0/24 and 4/24, Ornith 1/24 and 4/24, Muse 0/24 and 0/24; Muse adapter collection is incomplete. The 192-slot denominator included 150 accepted judgments, 19 budget failures and 23 parser failures; 17 total fully supported answers. The result is a tie among three adapters, not a factual winner. The sixteen current training runs establish token fit only; full per-epoch tables remain in Appendices E and F.

September 16–18 Muse experiments used previous methods. Their values do not enter current tables or charts. The eight corrected architecture runs supersede failed pre-fix attempts whose metrics are not active comparison evidence.

N=1,600 Family	Final Step	Selected Step	Final Train Loss / Token Accuracy	Selected Val. Loss / Token Accuracy
Gemma	600	400	0.2435 / 93.05%	0.5750 / 85.88%
Muse	600	200	0.0877 / 97.56%	0.5787 / 85.54%
Nemotron	1200	400	0.2265 / 92.98%	0.5527 / 85.56%
Ornith	1200	400	0.1796 / 94.01%	0.5626 / 85.38%

All four families used the same 124 validation prompt IDs and no protected test/control rows. Selected checkpoints were chosen within each run by validation loss. Nemotron and Ornith used early stopping (patience two, minimum improvement 0.005); all eight corrected archives passed 28/28 remote-local file-hash checks per run. Different tokenizers, recipes and response masks prevent treating the four validation-loss values as a shared factual score. The original thirty student-authored lookup controls are source-verified but inadequate for policy reasoning or abstention.

5. Selection and Justification

Nemotron 3.5 Lightning N=1,600 checkpoint-400 is the user-approved proposed candidate for a governed evaluation, not a proven winner or deployment choice. The October 8 advisory v2 probe produced a tie at 4/24 fully supported answers for Nemotron, Gemma and Ornith adapters; Nemotron was proposed because its 48 generations completed and it aligned with submitted human error-review scope, while retaining 19/24 known unsupported adapter outputs. Advisory scoring is not human factuality. The final fresh benchmark, source-exposure clearance, immutable seal, native matched-base proof, final run and genuinely new required ratings remain open. The historical test cannot be reused as untouched.

6. Interpretability and Leakage Check

The practical interpretability check for a language model is whether answers depend on the right exception, effective date, negation and source locator—not merely whether it emits a URL. No paired ablation has been run. Structural checks found zero document overlap, two duplicate-question groups with no cross-split match, one train–validation near-duplicate question candidate at Jaccard ≥0.85, and sixteen cross-split duplicate-answer groups affecting fifty-seven records. The group/record counts are reproduced from the canonical file with the project's punctuation-insensitive normalization; Appendix C lists each group. Identical short codes may be coincidental; repeated substantive policy text or template language may transmit information across split boundaries. Neither possibility is a human disposition.

Open Leakage and Quality Gate

Batch A human review was submitted October 5–6: all 16 cross-split answer groups (57 records) and the near-duplicate pair received human dispositions; the H2011 "Daily Max = 4" correction was recorded. This closes the review action, not every downstream remedy: a versioned re-split/re-hash and re-evaluation are still required before affected records can support comparative claims. The October 14 audit found no project-relative exposure evidence for 11 of 12 source documents and flagged one pharmacy source for review; unmatched bytes do not establish unseen policy semantics; final approval is false.

Every existing control item maps by full source URL and SHA-256 to a historical test document; Appendix B makes the per-item result visible. That refutes the specific concern that these thirty questions draw on training documents in the clean lineage, but does not fix their simple wording, source-page concentration, possible generated-test document overlap or absence of unanswerable cases. Preserve the original controls; the authorized separate local-AI benchmark must cover reasoning, dates, exceptions, conflicts and a defensible unsupported stratum, excluded from training and tuning. Its different authorship does not satisfy an independent-student-authorship requirement by itself; score abstention and locator correctness with explicit denominators and a human adjudication key. Do not call the old 1/1 safety probe a 90% estimate.

Attribution Is a Later Experiment

Open WebUI and ChromaDB retrieval have not been built and deployed for this comparison. No retrieval benefit is claimed. Model-only control scoring can follow a legitimate selection and dataset lock; paired retrieval-on versus retrieval-off attribution waits for the later deployed user interface. This dependency does not excuse unresolved leakage or missing model-only policy-support evidence.

7. What Would Change My Mind

Before final exposure, development evidence could justify candidate reconsideration. After final exposure, threshold failure withholds promotion and requires a separately governed later development cycle—not tuning or testing a rival on that same final cohort. The fixed candidate is rejected for promotion if source-verified policy questions do not improve answerable factual accuracy by at least 10 percentage points against untouched base, locator correctness is under 95%, unsupported claims reach 2% or more, abstention is under 90% on a meaningful unsupported set, safety routing is under 95%, general/format regression exceeds 3 points, interactive latency fails, or any leakage or rights defect persists. These are project targets, not results. A strong next-token score or a properly formatted URL cannot override human-scored factual and safety failures.

The October 14 decision supersedes the older October 9 fallback: compare fixed Nemotron N=1,600 checkpoint-400 with its matched untouched base using identical official-source passages and questions. This replaces the earlier Muse comparator and question-only input mode. It isolates adaptation given the same evidence, not retrieval quality or actual analyst practice. No additional training or candidate search is authorized here.

8. Repository and Reproducibility

Public course deliverables: github.com/porespellar/INAI6986_project_deliverables

Current evidence-replay notebook (not a final-test result): current_evidence_replay_20261016.ipynb

The current evidence-replay notebook recomputes the historical corpus split and overlap, all sixteen training summaries (28 Gemma/Muse and 24 Nemotron/Ornith epoch rows), and the 192-slot development advisory matrix from per-output records. Appendices E–F contain the epoch tables. It executes offline from a fingerprint-verified portable bundle; it does not rerun training, generation or final evaluation. The earlier notebook is retained only as a historical audit. No final factual metric can be replayed because no final run exists. Detailed private records remain in the source-of-record repository; instructor access must be confirmed separately.

Evidence Gates Before Comparative Scoring

Gate	Current Disposition / Required Action
Source membership	Closed for original 30: 30/30 on historical test documents; zero on clean train/validation. Preserve hash-matched item mapping.
Question validity	Open: create the approved local-AI/source-verified reasoning cohort. Original 30 remain unchanged; authorship limitation is disclosed.
Abstention	Open: verify a meaningful unsupported stratum against exact provided context; nine historical AI-assisted submissions are not the fresh cohort.
Duplicate/quality flags	Human Batch A dispositions submitted: 16 answer groups / 57 records + one near-duplicate pair; H2011 ruling: correct Daily Max to 4 in a new version; application to data/weights is not evidenced. Version remedies before reuse.
Candidate choice	Fixed for evaluation: Nemotron N1,600 checkpoint-400; selection is not a demonstrated factual win. Do not switch after final exposure.
Final model test	Not run: lock dataset, recipe, checkpoint, comparator and scorer first; evaluate the frozen candidate and untouched base once.

This supplement addresses feedback with verified human review, a disclosed provisional candidate proposal and an explicit unfinished-work boundary. It is not a claim that source exposure is approved, a clean expanded split exists, the final comparison has run or a particular course score has been achieved.

Appendix A — AI Use Disclosure

Hermes Agent (GPT-6 Luna draft; GPT-6.1 Sol review) and local Qwen assisted writing, code and evidence checks; local workers generated development data and advice. I designed and directed the project, authored and source-checked the original controls, supplied human ratings and retain responsibility. AI did not supply human ratings or final-test results; I must approve this draft before submission.

Expanded Assistance And Accountability Record

AI assistance. Hermes Agent (GPT-6 Luna 900k coordinating this document pass, with local Qwen3.8-Flash-Next-NVFP4 support for the Oct 16 evidence rebuild) reviewed course feedback, rubric, weekly instructions, training artifacts and layout; drafted and checked this update and its code. GPT-6.1 Sol 900k personally reviewed and corrected the evidence, narrative and final layouts in this pass. Local model workers—not the chat assistant—generated the original supervised data. AI prose does not count as human duplicate disposition, independently authored new control questions or measured factual outcomes.

Human authorship and accountability. J.O. conceived the project and selected the architecture, source scope, model candidates, metrics, split and guarded workflow; configured local hardware and services, directed agents, diagnosed training and checkpoint failures, and made proceed/stop/quarantine decisions. J.O. wrote and source-verified the original thirty control questions and answer keys. Nine supplementary pairs were AI-assisted/student-adopted; separately approved fresh benchmark wording will be local-AI-authored and source-verified, not independently student-authored. No additional student writing is requested; genuinely new output ratings and approval of conclusions remain required. The student retains final responsibility for all content and project actions.

Verification. The builder asserts the clean dataset hash, all 30 historical test-document mappings, the 16-group/57-record duplicate count, 16 completed nested-size training runs, 28 Gemma/Muse plus 24 Nemotron/Ornith epoch records, shared validation prompt identity and zero protected test/control use. The eight corrected architecture archives each match 28 remote-local artifact hashes. DOCX/PDF checks are delivery checks, not human adjudication.

Appendix B — Independent Control Source-Split Map

The original source-verified, student-authored set contains thirty answerable lookup questions from ten PDF documents and eleven document/page locations. The three most concentrated pages account for 21/30. Mapping uses full source URL plus SHA-256 against the canonical f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c dataset; every listed source is in the historical test split. The original questions are preserved verbatim; their source screenshot records remain in the original submission and local evidence directory. The table is a membership audit, not an expanded control or a model score.

ID	Original Student-Authored Question	Expected Answer	Source PDF / Page	Split
SA-001	What page number in the table of contents lists Rule 560-X-1-.07, Provider Rights and Responsibilities?	Page 3.	9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1	test
SA-002	What page number in the table of contents lists Rule 560-X-28-.02, Summary of State Law Regarding Advanced Directives?	Page 2.	9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1	test
SA-003	What page number in the table of contents lists Rule 560-X-49-.07, Copayment?	Page 2.	9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1	test
SA-004	What page number in the table of contents lists Rule 560-X-50-.03, Recipient Eligibility?	Page 3.	9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1	test
SA-005	What Medicaid provider type is assigned to Licensed Social Workers?	Provider type 11, Crossovers Only.	Apr26_23.pdf p.1	test
SA-006	What provider type is assigned to therapy providers?	Provider type 17, Therapy.	Apr26_37.pdf p.2	test
SA-007	What provider type is assigned to Remote Patient Monitoring providers?	Provider type 08, Remote Patient Monitoring.	Jul26_111.pdf p.1	test
SA-008	What provider type is assigned to physicians?	Provider type 31, Physician.	Jul26_28.pdf p.2	test
SA-009	What provider type is assigned to preventive health education providers?	Provider type 55.	Jul26_30.pdf p.1	test
SA-010	What provider type is assigned to psychology providers?	Provider type 54, Psychologist.	Jul26_34.pdf p.2	test
SA-011	What provider type is assigned to Behavioral Health providers?	Provider type 07, Behavioral Health.	Jul26_34.pdf p.2	test
SA-012	What specialty code is listed for Psychology providers?	Psychology specialty code 112.	Jul26_34.pdf p.2	test
SA-013	What specialty code is listed for a Licensed Professional Counselor / LPC?	Specialty code 070	Jul26_34.pdf p.2	test
SA-014	What provider type is assigned to Nurse Practitioners?	Provider type 09.	Jul26_28.pdf p.2	test
SA-015	What provider type is assigned to Physician Assistants?	Provider type 10.	Jul26_28.pdf p.2	test

Control Source-Split Map — Continued

ID	Original Student-Authored Question	Expected Answer	Source PDF / Page	Split
SA-016	What specialty code is listed for Occupational Therapy?	Occupational Therapy code 171.	Apr26_37.pdf p.2	test
SA-017	What specialty code is listed for Physical Therapy?	Physical Therapy code 170.	Apr26_37.pdf p.2	test
SA-018	What specialty code is listed for Speech Therapy?	Speech Therapy code 173.	Apr26_37.pdf p.2	test
SA-019	What specialty code is listed for Applied Behavior Analysis Therapy.	Applied Behavior Analysis (ABA) Therapy. Code 175.	Apr26_37.pdf p.2	test
SA-020	What provider specialties are assigned to Remote Patient Monitoring providers?	Provider specialties 085 or 086.	Jul26_111.pdf p.1	test
SA-021	What specialty code is listed for an Associate Licensed Counselor / ALC?	Specialty code 071.	Jul26_34.pdf p.2	test
SA-022	What specialty code is listed for a Licensed Marriage and Family Therapist / LMFT?	Specialty code 072.	Jul26_34.pdf p.2	test
SA-023	What specialty code is listed for a Licensed Marriage and Family Therapist Associate / LMFTA?	Specialty code 075.	Jul26_34.pdf p.2	test
SA-024	What specialty code is listed for a Licensed Master Social Worker / LMSW?	Specialty code 073.	Jul26_34.pdf p.2	test
SA-025	What specialty code is listed for a Licensed Independent Clinical Social Worker / LICSW.	Specialty code 074.	Jul26_34.pdf p.2	test
SA-026	What specialty code is listed for a Licensed Psychological Technician / LPT?	Specialty code 076.	Jul26_34.pdf p.2	test
SA-027	What specialty code is listed for Preventive Health Education?	Specialty code 183.	Jul26_30.pdf p.1	test
SA-028	What specialty code is listed for EPSDT in the physician specialty table?	EPSDT code 560.	Jul26_28.pdf p.2	test
SA-029	What specialty code is listed for Family Practice in the physician specialty table?	Family Practice code 316.	Jul26_28.pdf p.2	test
SA-030	What specialty code is listed for Cardiac Surgery in the physician specialty table?	Cardiac Surgery code 312.	Jul26_28.pdf p.2	test

Source concentration (document/page → number of questions): Jul26_34.pdf p.2: 10; Jul26_28.pdf p.2: 6; Apr26_37.pdf p.2: 5; Jul26_111.pdf p.1: 2; Jul26_30.pdf p.1: 2; 9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1: 1; 9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1: 1; 9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1: 1; 9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1: 1; Apr26_23.pdf p.1: 1.

Appendix C — Cross-Split Answer-Overlap Audit

These are reproducible flags, not human rulings. The canonical file yields sixteen punctuation/case-normalized answer groups spanning splits (57 member records). Batch A has already supplied dispositions for all groups; a shared page number, repeated boilerplate and repeated policy answer require different remedies. Do not repeat that human work. Do not delete records merely because an answer string repeats. Keep the frozen original file and version a corrected split/file after human approval.

Answer Group (abbreviated)	Members	Splits	Provisional Risk / Action
page 2	8	test/train/validation	Batch A ruling received; remedy not evidenced
this page intentionally left blank	7	test/train/validation	Batch A ruling received; remedy not evidenced
the current procedural terminology cpt and current dental termin…	6	train/validation	Batch A ruling received; remedy not evidenced
page 3	5	test/train/validation	Batch A ruling received; remedy not evidenced
3 90	4	test/train	Batch A ruling received; remedy not evidenced
icd 9 codes should be used for claims submitted with dates of se…	4	test/train/validation	Batch A ruling received; remedy not evidenced
3	3	test/train	Batch A ruling received; remedy not evidenced
4	3	train/validation	Batch A ruling received; remedy not evidenced
chapter 5	3	train/validation	Batch A ruling received; remedy not evidenced
11	2	test/train	Batch A ruling received; remedy not evidenced
334 215 7416	2	test/train	Batch A ruling received; remedy not evidenced
a hard copy ub 04 claim form must be submitted	2	train/validation	Batch A ruling received; remedy not evidenced
the amount billed shall not exceed the entity s actual acquisiti…	2	test/train	Batch A ruling received; remedy not evidenced
the current procedural terminology cpt and current dental termin…	2	train/validation	Batch A ruling received; remedy not evidenced
the ltc gainwell cover sheet from the web portal must accompany …	2	train/validation	Batch A ruling received; remedy not evidenced
up to four procedure code modifiers	2	test/train	Batch A ruling received; remedy not evidenced

The cross-split near-duplicate pair also has a Batch A human disposition; any required versioned remedy and its effect on selected weights remain unevidenced. The separate quality-review human template and AI example are in artifacts/milestone1/quality_review/2026-09-24/; no sign-off is inferred from their existence. The original generated-set thirty-card human factual pilot is a different activity.

Appendix D — Local Evidence Ledger and Sources

Claim / Requirement	Verifiable Artifact
Rubric and instructor's scored feedback	Assignment 3: Milestone 1 — Data & Model Selection, course rubric; instructor's scored feedback (private course file)
Historical notebook and original course record	notebook/milestone1_reproducibility.ipynb; artifacts/milestone1/2026-09-22/{progress_snapshot,evaluation_snapshot}.json; original v2 PDF
Clean dataset and frozen control	artifacts/milestone1/2026-09-17/approved_candidates.jsonl (SHA-256 f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c); data/curated/student-authored-control-set-2026-09-22.jsonl (SHA-256 a1deb52304d7bce939593a228836debe2359488159e6e3b112e9d48ac7bea55c)
Human pilot and quality flags (historical archive only)	artifacts/milestone1/human_scoring/2026-09-24/human_generated_sample_scores.json; artifacts/milestone1/quality_review/2026-09-24/INAI6986_1999Record_QualityReview_HumanTemplate.xlsx; these are not current matched-run metrics.
Gemma/Muse current training cohort	artifacts/model_comparison/2026-09-25/results.jsonl (28 epoch records); Appendix E; eight matched nested-size runs.
Gemma and Muse per-run completion artifacts	artifacts/model_comparison/2026-09-25/training_results/ and muse-runs/; all final states verified complete, best checkpoints selected by validation loss, no test/control rows used.
Legacy Muse provenance (excluded)	artifacts/model_comparison/2026-09-25/historical-run-results.json and legacy-muse-comparability-note.json; retained only as history and not included in current results tables/plots.
Corrected Nemotron/Ornith cohort	artifacts/model_comparison/2026-09-25/architecture-training-gates/clean-runs/architecture-clean-run-matrix.json (24 epoch records, eight run archives); Appendix F; run_log.csv totals sixteen runs. Superseded failed attempts excluded.

Selected Technical Sources

Alabama Medicaid Administrative Code:
https://medicaid.alabama.gov/content/9.0_Resources/9.2_Administrative_Code.aspx

Alabama Medicaid Provider Manuals:
https://medicaid.alabama.gov/content/Gated/7.6.1G_Provider_Manuals.aspx

LoRA method: Hu et al., https://arxiv.org/abs/2106.09685. Candidate model identities and exact pinned revisions are recorded in artifacts/model_comparison/2026-09-25/source-state-inventory.json; no third-party performance claim is inferred from a model card.

Appendix E — Complete Current Gemma and Muse Epoch Results

This table contains the 28 current training/validation records from results.jsonl and excludes the pre-amendment September 16–18 Muse runs. The eight runs use document-balanced nested train prompt IDs (N=200/509/750/1,600) and the same fixed 124 validation IDs for both model families. Results are teacher-forced causal next-token metrics only. A dash means that metric was not recorded for the checkpoint, not zero; for example, the partial-epoch Gemma N=509 step-200 checkpoint has validation but no train-split metric. The step-200 Gemma N=509 row is a partial-epoch work-window checkpoint before the run resumed; the final completed run ended at step 320.

Model / N	Epoch / Step	Train Loss	Train Next-Token Acc.	Val. Loss	Val. Next-Token Acc.	Scored Tokens: Train / Val.
Gemma 4; N=200	1.000 / 25	1.1405	74.60%	—	—	— / —
Gemma 4; N=200	2.000 / 50	0.7575	82.99%	0.7751	82.15%	28,056 / 18,125
Gemma 4; N=200	3.000 / 75	0.6893	84.40%	0.7044	83.45%	28,056 / 18,125
Gemma 4; N=509	1.000 / 64	0.6592	84.73%	0.6508	84.19%	73,166 / 18,125
Gemma 4; N=509	2.000 / 128	0.5430	86.60%	0.6060	84.93%	73,166 / 18,125
Gemma 4; N=509	3.000 / 192	0.4433	88.52%	0.6067	85.19%	73,166 / 18,125
Gemma 4; N=509	3.126 / 200	—	—	0.6178	85.25%	— / 18,125
Gemma 4; N=509	4.000 / 256	0.3688	90.06%	0.6321	85.24%	73,166 / 18,125
Gemma 4; N=509	5.000 / 320	0.3373	90.81%	0.6636	85.03%	73,166 / 18,125
Gemma 4; N=750	1.000 / 94	0.6182	85.26%	0.6199	84.68%	109,023 / 18,125
Gemma 4; N=750	2.000 / 188	0.4791	87.69%	0.5829	85.58%	109,023 / 18,125
Gemma 4; N=750	3.000 / 282	0.3361	90.63%	0.6205	85.37%	109,023 / 18,125
Gemma 4; N=750	4.000 / 376	0.2192	93.62%	0.6946	85.13%	109,023 / 18,125
Gemma 4; N=1,600	1.000 / 200	0.5400	86.69%	0.5751	85.79%	232,550 / 18,125
Gemma 4; N=1,600	2.000 / 400	0.3832	89.63%	0.5750	85.88%	232,550 / 18,125
Gemma 4; N=1,600	3.000 / 600	0.2435	93.05%	0.6180	85.65%	232,550 / 18,125
Muse Glimmer; N=200	1.000 / 25	0.5252	86.63%	0.5853	85.10%	24,507 / 15,903
Muse Glimmer; N=200	2.000 / 50	0.3505	90.32%	0.6073	84.98%	24,507 / 15,903
Muse Glimmer; N=200	3.000 / 75	0.2376	93.15%	0.6589	84.54%	24,507 / 15,903
Muse Glimmer; N=509	1.000 / 64	0.4638	87.86%	0.5727	85.44%	64,019 / 15,903
Muse Glimmer; N=509	2.000 / 128	0.2305	93.62%	0.6264	84.86%	64,019 / 15,903
Muse Glimmer; N=509	3.000 / 192	0.0856	97.64%	0.7456	84.70%	64,019 / 15,903
Muse Glimmer; N=750	1.000 / 94	0.4532	88.04%	0.5725	85.44%	95,431 / 15,903
Muse Glimmer; N=750	2.000 / 188	0.2137	93.82%	0.6279	85.32%	95,431 / 15,903
Muse Glimmer; N=750	3.000 / 282	0.0846	97.70%	0.7447	84.80%	95,431 / 15,903
Muse Glimmer; N=1,600	1.000 / 200	0.4067	89.15%	0.5787	85.54%	203,025 / 15,903
Muse Glimmer; N=1,600	2.000 / 400	0.1969	94.23%	0.6054	85.41%	203,025 / 15,903
Muse Glimmer; N=1,600	3.000 / 600	0.0877	97.56%	0.7249	84.72%	203,025 / 15,903

Appendix F — Corrected Nemotron and Ornith Epoch Results

Twenty-four epoch records from the eight corrected October 1 Nemotron/Ornith runs. The per-run run_config, training_summary, trainer/final state, selected/final checkpoints, best adapter, logs and integrity manifest remain in the private project archive; 28/28 remote-local hashes matched per run. This is teacher-forced token fit, not factual-answer accuracy. Pre-fix failed attempts are excluded.

Model / N	Epoch / Step	Train Loss	Train Next-Token Acc.	Val. Loss	Val. Next-Token Acc.	Scored Tokens: Train / Val.
Nemotron; N=200	1 / 50	0.5403	85.57%	0.5859	84.73%	27,126 / 17,335
Nemotron; N=200	2 / 100	0.4232	88.02%	0.5825	84.75%	27,126 / 17,335
Nemotron; N=200	3 / 150	0.3895	88.65%	0.5862	84.82%	27,126 / 17,335
Nemotron; N=509	1 / 128	0.4874	86.59%	0.5614	85.25%	70,628 / 17,335
Nemotron; N=509	2 / 256	0.3613	89.43%	0.5637	85.30%	70,628 / 17,335
Nemotron; N=509	3 / 384	0.2429	92.52%	0.6118	84.83%	70,628 / 17,335
Nemotron; N=750	1 / 188	0.4774	86.66%	0.5574	85.23%	105,256 / 17,335
Nemotron; N=750	2 / 376	0.3577	89.46%	0.5593	85.37%	105,256 / 17,335
Nemotron; N=750	3 / 564	0.2363	92.61%	0.6060	85.17%	105,256 / 17,335
Nemotron; N=1,600	1 / 400	0.4671	87.09%	0.5527	85.56%	224,733 / 17,335
Nemotron; N=1,600	2 / 800	0.3509	89.76%	0.5549	85.49%	224,733 / 17,335
Nemotron; N=1,600	3 / 1200	0.2265	92.98%	0.6001	85.15%	224,733 / 17,335
Ornith; N=200	1 / 50	0.5132	86.06%	0.6011	84.17%	26,805 / 17,279
Ornith; N=200	2 / 100	0.3855	88.72%	0.6028	84.62%	26,805 / 17,279
Ornith; N=200	3 / 150	0.3454	89.65%	0.6184	84.57%	26,805 / 17,279
Ornith; N=509	1 / 128	0.4750	86.99%	0.5822	84.96%	69,820 / 17,279
Ornith; N=509	2 / 256	0.3272	90.14%	0.5829	84.85%	69,820 / 17,279
Ornith; N=509	3 / 384	0.1945	93.56%	0.6390	84.89%	69,820 / 17,279
Ornith; N=750	1 / 188	0.4683	86.92%	0.5809	84.98%	104,069 / 17,279
Ornith; N=750	2 / 376	0.3251	90.16%	0.5834	85.17%	104,069 / 17,279
Ornith; N=750	3 / 564	0.1857	93.92%	0.6355	84.68%	104,069 / 17,279
Ornith; N=1,600	1 / 400	0.4613	87.24%	0.5626	85.38%	222,040 / 17,279
Ornith; N=1,600	2 / 800	0.3083	90.62%	0.5797	85.46%	222,040 / 17,279
Ornith; N=1,600	3 / 1200	0.1796	94.01%	0.6254	85.21%	222,040 / 17,279

Within-family checkpoint selection is based on validation loss; no global candidate or deployment winner is asserted. The fixed validation identity is bound in architecture-clean-run-matrix.json; no test or independent-control rows entered these runs.

Appendix G — Instructor Feedback Response

Every substantive feedback item is mapped to its verified response and remaining limitation. Completed human work is preserved; new local-AI wording is not called student-authored. No grade or waiver is assumed.

ID	Feedback Concern	Response, Evidence And Remaining Limitation
M1-01	Lookup-only questions	Preserve all 30 originals; separately authorized reasoning/date/exception/conflict benchmark is not yet created. Local-AI authorship is disclosed; no instructor waiver.
M1-02	Unsupported questions	Nine historical supplements received: eight adopted AI drafts, one edited. They are not independent student authorship or a verified fresh unsupported stratum.
M1-03	Narrow source coverage	All 30 controls map by URL and hash to historical-test sources; 21/30 on three pages. Wording independence does not imply unseen sources.
M1-04	Ceiling baseline / URL formatting	Manual 30/30 is an easy lookup reference; 84 seconds is a retrospective proxy. Current 192-slot advisory results are not final/human accuracy or time savings.
M1-05	Duplicate-answer leakage	All 16 groups / 57 records have Batch A human dispositions. Application of versioned remedies to canonical data and selected weights is not evidenced.
M1-06	Insufficient alternatives	Four genuine families at four nested sizes; 16 completed runs. Three adapters tie at 4/24 advisory support. Fixed Nemotron400 is a governed candidate, not a winner.
M1-07	Overlong control appendix	Two compact source-split tables preserve original wording and show coverage; original detailed evidence remains archived.
M1-08	Source-column and exposure defects	Near-duplicate ruling and H2011 Daily Max=4 correction direction received. Do not claim existing weights were repaired. Historical test remains exposed.
M1-09	Priority before next milestone	Completed review/refinement work is preserved. Source-only recovery integration passed; fresh benchmark, native parity, seal, final outputs and new human ratings remain missing.

Appendix H — Rubric Evidence And Credit Rationale

This crosswalk requests credit only for work evidenced here. A criterion with an open gate is identified rather than treated as satisfied; the appendix is not an instruction to the grader or a score guarantee.

M1 Criterion	Evidence in this report	Defensible claim / remaining limitation
Data Preparation & Split (20)	Section 1, p.1; Appendices B–D, pp.7–10; canonical data hash and 30/30 source-to-historical-test map.	Document-grouped 1,684/124/191 split and leakage findings are reproducible; fresh source exposure and corrected split remain unresolved.
Baseline (15)	Section 3, p.2; Section 4, p.3; current evidence-replay notebook.	Manual 30/30 lookup and retrospective timing proxy are clearly bounded. Matched-model base exists as a planned technical comparator, not an executed real-practice comparison.
Candidate Reasoning (20)	Sections 2, 4–5, pp.2–4; Appendices E–F.	Four trained families at four sizes; provisional Nemotron proposal follows development evidence. No final factual winner or deployment promotion is claimed.
Evidence & Evaluation (20)	Section 4; actual 192-slot advisory counts; Appendices E–F.	Training and development denominators are explicit. Advisory scores are not human/final metrics; same-split final scoring remains open.
Interpretability & Leakage (15)	Section 6, p.4; Appendices B–C; Appendix G feedback response.	Document split, answer overlap, source-map and human dispositions are reported. Remedies need versioned re-split; no ablation or final source-exposure clearance.
Reasoning Quality & Format (10)	Sections 5, 7–8; AI disclosure; notebook status.	Promotion and falsification thresholds and limitations are explicit. Current development replay executes; final factual experiment and its raw-to-metric notebook remain incomplete.